Question 5 - Set up version control using Git

Question 5: Initialize a repository, stage and commit changes, create branches, merge branches, and push the repository to GitHub.

Repository: https://github.com/ksamirdev/ensp461-devops-lab1

Workflow

```
$ git init
Initialized empty Git repository in /home/samir/ensp461-devops-lab1/.git/

$ git status --short
?? .gitignore
?? README.md
?? requirements.txt
?? service/

$ git add .

$ git commit -m 'Add text statistics Flask service'
[main (root-commit) 7b5d99f] Add text statistics Flask service
 5 files changed, 63 insertions(+)
 create mode 100644 .gitignore
 create mode 100644 README.md
 create mode 100644 requirements.txt
 create mode 100644 service/__init__.py
 create mode 100644 service/web.py

$ git checkout -b feature/test-suite
Switched to a new branch 'feature/test-suite'

$ git add tests

$ git commit -m 'Add pytest suite for service endpoints'
[feature/test-suite 9c2335f] Add pytest suite for service endpoints
 1 file changed, 29 insertions(+)
 create mode 100644 tests/test_web.py

$ python3 -m venv .venv && .venv/bin/pip install -q -r requirements.txt && .venv/bin/python -m pytest -v
============================= test session starts ==============================
platform linux -- Python 3.12.3, pytest-8.3.4, pluggy-1.7.0 -- /home/samir/ensp461-devops-
lab1/.venv/bin/python
cachedir: .pytest_cache
rootdir: /home/samir/ensp461-devops-lab1
collecting ... collected 5 items

tests/test_web.py::test_healthz_reports_healthy PASSED                   [ 20%]
tests/test_web.py::test_text_stats_counts_unique_words_case_insensitively PASSED [ 40%]
tests/test_web.py::test_stats_endpoint_returns_counts PASSED             [ 60%]
tests/test_web.py::test_stats_endpoint_rejects_invalid_text[body0] PASSED [ 80%]
tests/test_web.py::test_stats_endpoint_rejects_invalid_text[body1] PASSED [100%]

============================== 5 passed in 0.05s ===============================

$ git checkout main
Switched to branch 'main'

$ git merge --no-ff feature/test-suite -m 'Merge feature/test-suite into main'
Merge made by the 'ort' strategy.
 tests/test_web.py | 29 ++++++++++++++++++++++++++++
 1 file changed, 29 insertions(+)
 create mode 100644 tests/test_web.py

$ git tag -a v1.0.0 -m 'First tested release'

$ git branch -a
  feature/test-suite
* main
```

Fig: init, stage, commit, branch, tests, merge, tag (rendered from evidence/logs/q5-git-workflow.log)

```
$ git push origin main --tags
To https://github.com/ksamirdev/ensp461-devops-lab1.git
   3d80611..8633054  main -> main

$ git push origin main --tags
To https://github.com/ksamirdev/ensp461-devops-lab1.git
   8633054..e93d90f  main -> main
 * [new tag]         v1.1.0 -> v1.1.0
```

Fig: Pushes to GitHub, run from the host machine (rendered from evidence/logs/q5-push.log)

Commit history

```
* 81a3761 2026-11-05 18:29 Add lab command logs, screenshots and tooling
* e93d90f 2026-11-05 17:41 Bump release to 1.1.0 to exercise automatic deployment
* 8633054 2026-11-05 17:39 Add declarative Jenkins pipeline with build, test, package and deploy stages
* a34c4f5 2026-11-05 17:39 Add systemd unit and health-checked deploy script
*   3d80611 2026-11-05 17:35 Merge feature/test-suite into main
|\
| * 9c2335f 2026-11-05 17:35 Add pytest suite for service endpoints
|/
* 7b5d99f 2026-11-05 17:35 Add text statistics Flask service
```

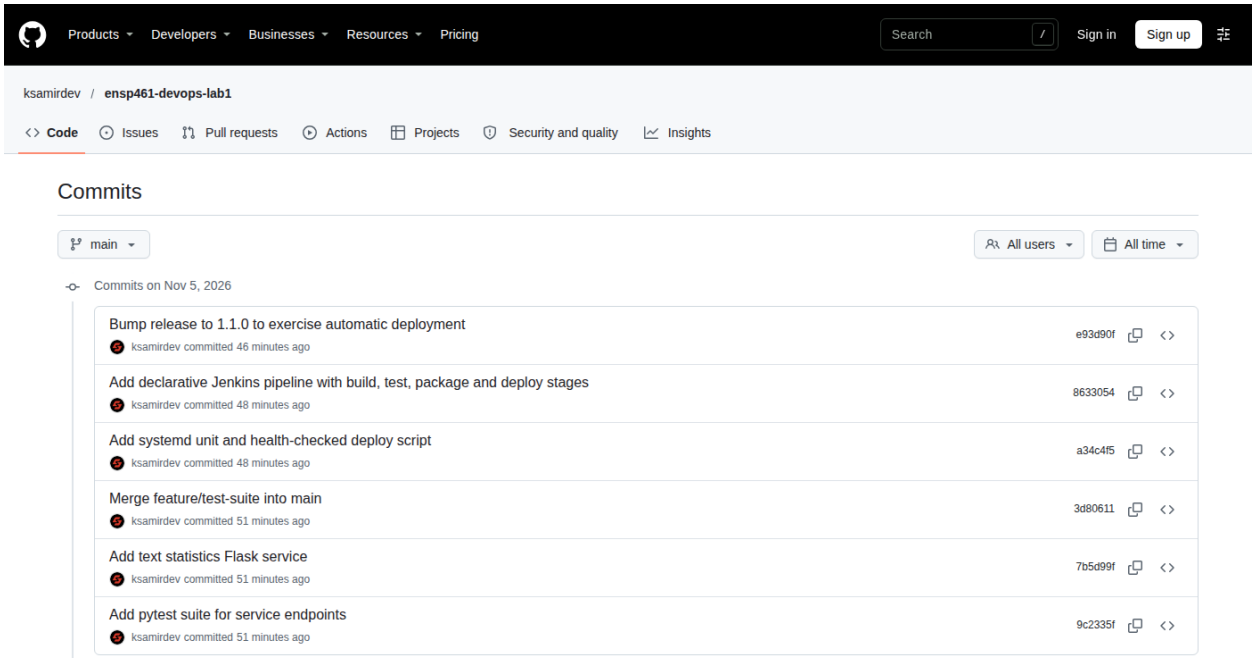


Fig: Commits on GitHub

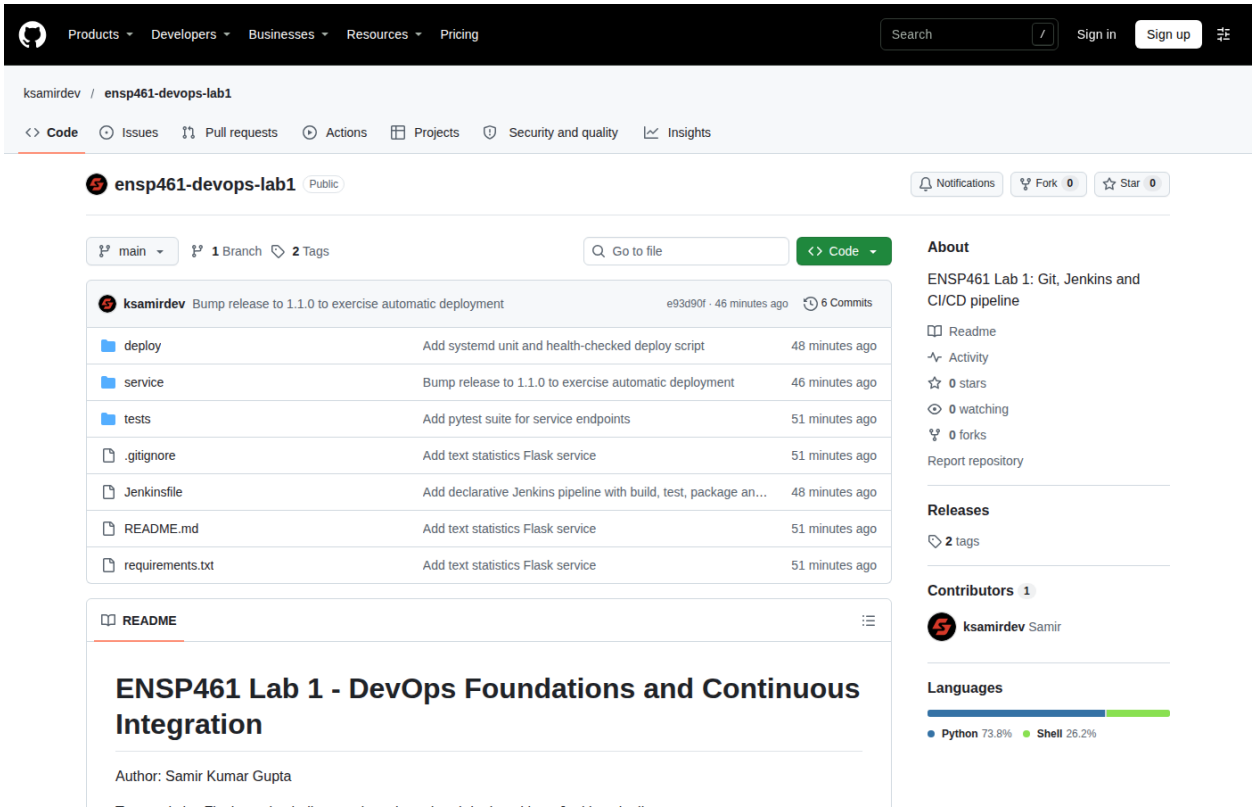


Fig: Public repository on GitHub

Observations

- The feature branch kept main unchanged until the test suite passed.
- Merging with --no-ff keeps a merge commit, so the graph shows when the feature landed.
- An annotated tag marks the first tested release, v1.0.0.
- Output of the very first push was not captured. The later push log and git ls-remote output confirm the branch and tag on GitHub.